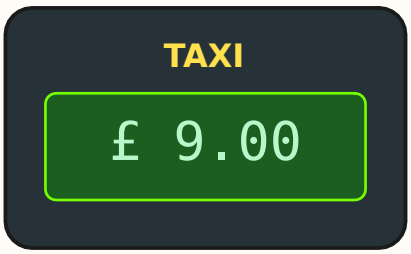
Chapter 20

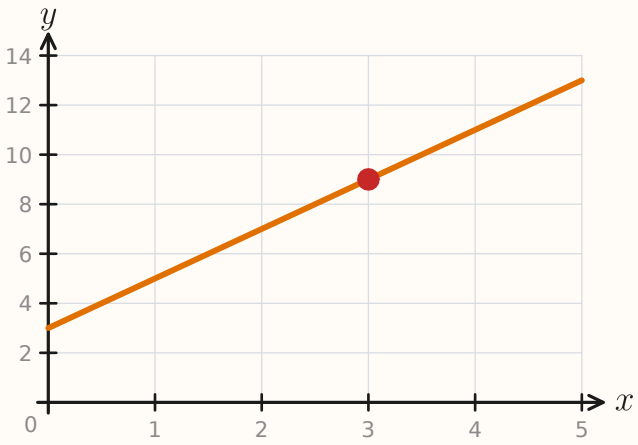
The straight line

y = mx + c, and everything you can be asked about it



£3 to get in, then £2 per mile.

3 miles: 3 + 2 × 3 = £9



Anything that grows by the **same amount for every step** (a taxi fare, a phone bill, a spring) graphs as a **straight line**, and every straight line (except vertical ones) has the formula

$$y = mx + c$$

This chapter states every rule you need about straight lines and then practises each type of question: reading m and c, finding the line through two points, **proving lines are parallel or perpendicular**, midpoints and distances, where lines cross, and **filling in a missing value** when you're only given part of the information.

Goes with **Lecture 7** (y = mx + c, slope and intercept) and **Lecture 14** (line through two points). Chapter 34 reuses this chapter for tangents and normals.

y = mx + c: gradient and intercept

RULE: the equation of a straight line

$y = mx + c$, where **m** is the **gradient** (how much y changes when x goes up by 1) and **c** is the **y-intercept** (the value of y where the line crosses the y-axis, i.e. at $x = 0$).

To read m and c, the equation must start "$y =$" with y on its own. If it doesn't, rearrange first.

A point (a, b) lies on the line exactly when putting $x = a$ into the equation gives $y = b$.

Fare in pounds y for x miles: $y = 2x + 3$. The table (like Lecture 7's for $y = 2x + 1$):

miles x	0		1		2		3		4		5
2x	0		2		4		6		8		10
+ 3	3		3		3		3		3		3
fare y	**3**	+2	**5**	+2	**7**	+2	**9**	+2	**11**	+2	**13**

The fares go up by **2 every time** the miles go up by 1: that's m. At 0 miles you still pay **3**: that's c.



c = 3: where the line crosses the y-axis (purple dot)

m = 2: every 1 step across, the line climbs 2 (the little staircase)

Gradient: rise over run

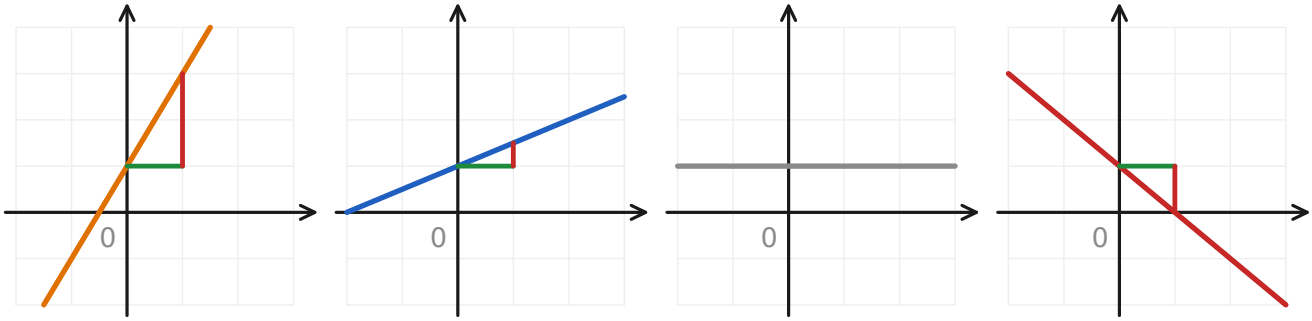
RULE: gradient

$m = \dfrac{\text{rise}}{\text{run}} = \dfrac{\text{change in } y}{\text{change in } x}$, read **left to right**.

$m > 0$: uphill. $m < 0$: downhill. $m = 0$: flat (horizontal).

Steepness depends on the **size** of m ignoring its sign, $|m|$: $y = -5x$ is steeper than $y = 2x$.

Road signs use this. "10%" means rise 10 m for every 100 m across: gradient $\frac{10}{100} = 0.1$. "1 in 4" means rise 1 for every 4 across: 0.25, a steep hill.



m = 2: steep uphill **m = 0.5: gentle uphill** **m = 0: flat (y = c)** **m = −1: downhill**

Lecture 7 says "the greater m, the steeper the line": strictly, it is the greater **size** of m. A negative m tilts the line the other way.

RULE: horizontal and vertical lines

Horizontal line through (a, b): $y = b$. Gradient 0.

Vertical line through (a, b): $x = a$. The run is 0 and you can't divide by 0, so the gradient is **undefined**. A vertical line can't be written as $y = mx + c$.

The x-axis is the line $y = 0$; the y-axis is the line $x = 0$.

The line through two points (Lecture 14)

RULE: gradient through two points

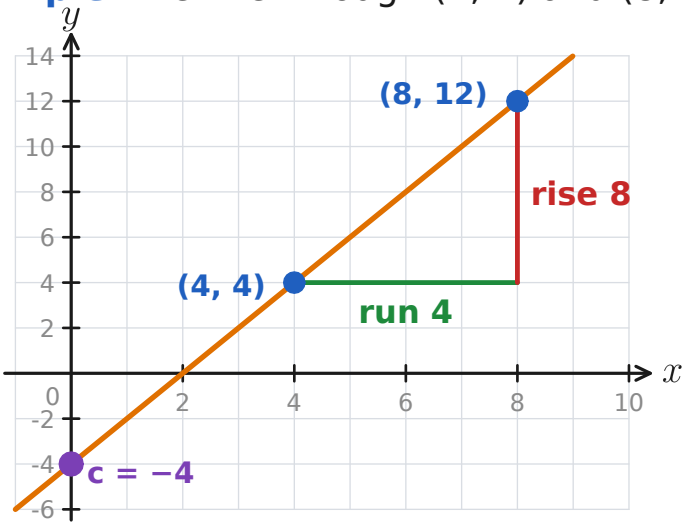
For points (x_1, y_1) and (x_2, y_2): $m = \dfrac{y_2 - y_1}{x_2 - x_1}$.

Same order top and bottom: second minus first in both, or first minus second in both.

PROCEDURE: equation of the line through two points

1. Find m with the gradient formula.
2. Put m into $y = mx + c$.
3. Substitute **either** point for x and y, and solve for c.
4. Write the final equation, and check it with the **other** point.

Example: the line through (4, 4) and (8, 12).



From (4, 4) to (8, 12):
across 8 − 4 = 4
up 12 − 4 = 8

The line crosses the y-axis at −4, below 0.

$$m = \frac{12 - 4}{8 - 4} = \frac{8}{4} = 2$$ ← *step 1: gradient formula*

$$y = 2x + c$$ ← *step 2*

$$4 = 2 \times 4 + c \;\Rightarrow\; c = -4$$ ← *step 3: use (4, 4)*

$$y = 2x - 4$$ ← *step 4. Check (8, 12): 2 × 8 − 4 = 12 ✓*

Point-gradient form: the fastest route

RULE: point-gradient form

The line with gradient m through the point (x_1, y_1) is $y - y_1 = m(x - x_1)$.

Use it whenever you know **one point and the gradient**. Expand the bracket and add y_1 to both sides to get $y = mx + c$.

Watch negative coordinates: through $(-3, -2)$ it is $y - (-2) = m(x - (-3))$, i.e. $y + 2 = m(x + 3)$.

It is the gradient formula rearranged: for any point (x, y) on the line, $\frac{y-y_1}{x-x_1} = m$. Multiply both sides by $(x - x_1)$.

Example: the same line through (4, 4) with gradient 2.

$$y - 4 = 2(x - 4)$$ ← point (4, 4), m = 2

$$y - 4 = 2x - 8$$ ← expand

$$y = 2x - 4$$ ← +4 both sides: same answer as before

Example: gradient $-\frac{1}{2}$ through $(-2, 3)$.

$$y - 3 = -\tfrac{1}{2}(x - (-2)) = -\tfrac{1}{2}(x + 2)$$

$$y = -\tfrac{1}{2}x - 1 + 3 = -\tfrac{1}{2}x + 2$$

A downhill example, in tiny steps

Line through (−1, 5) and (3, −3). The minus signs are where mistakes happen, so every subtraction gets its own bracket:

$$m = \frac{(-3) - (5)}{(3) - (-1)}$$ ← second minus first, top and bottom

$$= \frac{-8}{4} = -2$$ ← −3 − 5 = −8; 3 − (−1) = 3 + 1 = 4

$$5 = -2 \times (-1) + c$$ ← use (−1, 5)

$$5 = 2 + c \;\Rightarrow\; c = 3$$ ← −2 × −1 = +2

$$y = -2x + 3$$

Check: $-2(3) + 3 = -3$ ✓. *Sense check:* from the first point to the second, x went up but y went down, so the gradient must be negative. It is.

Lines in disguise, and the cover-up sketch

RULE: other forms of a line

A line may be written $ax + by = d$ or $ax + by + c = 0$. To find its gradient and intercept, **rearrange to** $y = mx + c$ (Chapter 14). Shortcut: for $ax + by = d$, $m = -\frac{a}{b}$.

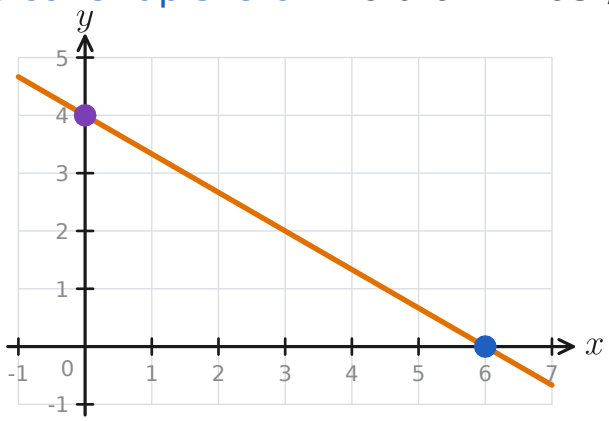
To write $y = mx + c$ as $ax + by + c = 0$ with whole numbers: multiply by the denominators, then move every term to one side.

Axis crossings: the line crosses the y-axis where $x = 0$ and the x-axis where $y = 0$.

$$2x + 3y = 12 \Rightarrow 3y = -2x + 12$$ ← −2x from both sides

$$y = -\frac{2}{3}x + 4$$ ← ÷ 3: so m = −2/3 and c = 4

The cover-up sketch. To draw it fast, find where it crosses each axis:



Cover up the x-term (x = 0):
3y = 12, so y = 4. Point (0, 4)

Cover up the y-term (y = 0):
2x = 12, so x = 6. Point (6, 0)

Join the two dots with a ruler.

Example: write $y = \frac{3}{4}x - 2$ in the form $ax + by + c = 0$.

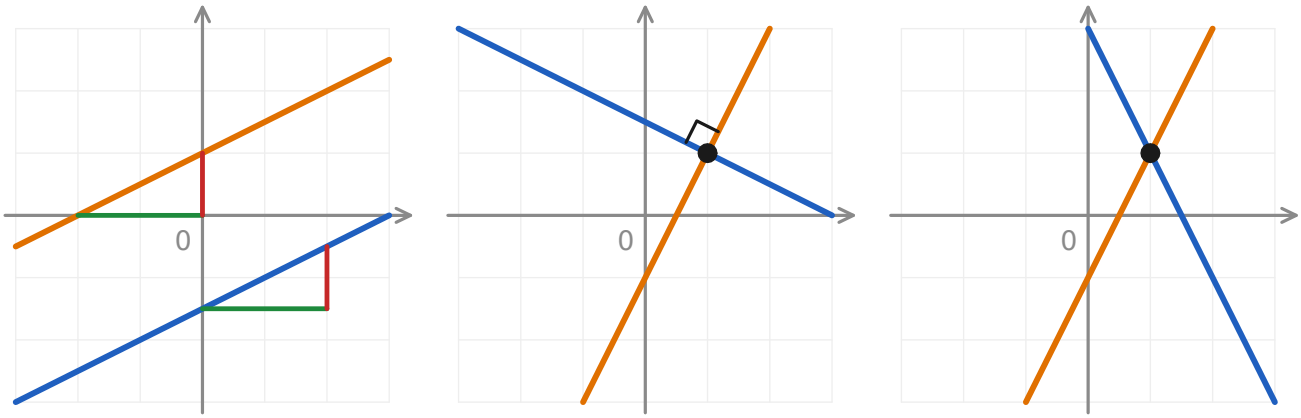
$$4y = 3x - 8$$ ← × 4 to clear the fraction

$$3x - 4y - 8 = 0$$ ← move everything to one side

Two lines at once: only three things can happen

Until now every question had **one** line. When a second line turns up there is only one new question: **how do the two lines sit next to each other?**

Remember what the two numbers do. m is the line's **tilt**. c only **slides** the line up or down without turning it. So to compare how two lines tilt, **ignore the c's and compare the m's**. There are only three possible results:



PARALLEL

same m, different c
same tilt: they never meet
(train tracks)

PERPENDICULAR

m's multiply to −1
meet at a square corner
(a + sign, a crossroads)

NEITHER

any other pair of m's
cross at a slant
(a squashed X)

The words, decoded

Word	Picture it as	What it means for the m's
Parallel	Train tracks, lanes on a motorway, the lines on lined paper. Look at the word itself: para**ll**el has two parallel l's in the middle.	Same tilt: $m_1 = m_2$. The c's are different (same m **and** same c is just one line drawn twice). They never meet.
Perpendicular (symbol ⊥)	The corner of a page, a + sign, a crossroads, a wall standing on the floor. The symbol ⊥ is exactly that: a wall on a floor. The word comes from the Latin for a builder's **plumb line**, a weight on a string that hangs straight down and so meets level ground at a perfect square corner.	They meet at a square corner (90°). Flip one gradient and change its sign to get the other: $m_2 = -\frac{1}{m_1}$.
Neither	Two roads that cross at a slant.	Any other pair of m's. They meet once, but not at a square corner.

Same meaning, different costume: "at right angles to", "normal to" (Chapter 34) and "orthogonal" all mean **perpendicular**.

Name tags. With two lines around, the working needs to say which is which. L_1 and L_2 just mean "line 1" and "line 2"; m_1 is "the gradient of line 1"; m_{AB} is "the gradient of the line from A to B". The small numbers and letters are labels, not something to calculate.

✎ YOUR GO (pencil first)

$y = 3x + 1$ and $y = 3x - 4$. Before any algebra: will these two lines ever meet? Explain using only the numbers in the equations.

Answer: no. Both have m = 3, so they tilt exactly the same way; c = 1 and c = −4 only put one 5 units above the other. They are parallel.

Parallel and perpendicular: the gradient rules

RULE: parallel and perpendicular gradients

Parallel lines have **equal gradients**: $m_1 = m_2$. (Same m and same c means it is the same line.)

Perpendicular lines (meeting at 90°) have gradients that **multiply to −1**: $m_1 \times m_2 = -1$.

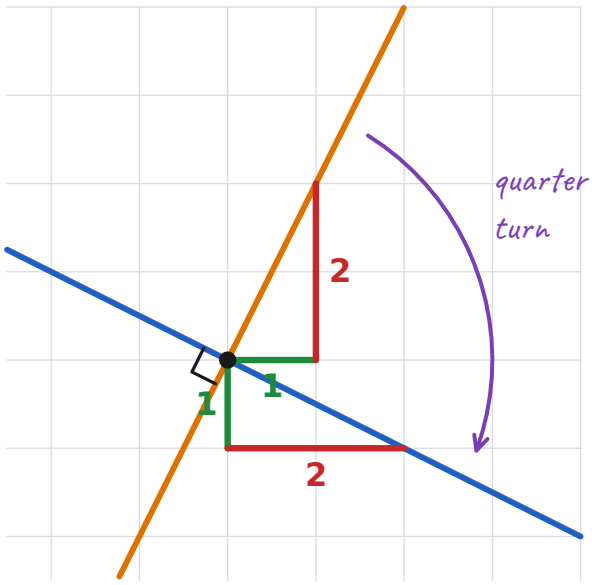
So the perpendicular gradient is the **negative reciprocal**: $m_2 = -\dfrac{1}{m_1}$.

Flip the fraction **and** change the sign: $3 \to -\frac{1}{3}$, $-\frac{2}{5} \to \frac{5}{2}$.

Exception: a horizontal line ($m = 0$) and a vertical line (no gradient) are perpendicular, but the product rule can't be used for them.

Why "flip and change the sign"? Give the staircase a quarter turn.

The orange line's staircase goes across 1, up 2, so $m = 2$. Now turn the whole staircase a quarter turn clockwise, like turning a page on its side, and watch its two sides:



Before: across 1, up 2.
Orange m = 2/1 = 2

After: across 2, down 1.
Blue m = −1/2

The 1 and the 2 swapped places: that's the FLIP.

Uphill turned into downhill: that's the MINUS sign.

So a perpendicular gradient is always the old one **flipped** (rise and run swap places) **with its sign changed** (uphill becomes downhill). Multiply the pair and the numbers cancel, leaving only the minus: $2 \times (-\frac{1}{2}) = -1$. That is all "$m_1 m_2 = -1$" is saying.

SENSE CHECK: does my perpendicular gradient look right?

A perpendicular pair always has **one uphill and one downhill** line (opposite signs), and **if one is steep the other is gentle** (one gradient bigger than 1 in size, the other smaller).

2 and $-\frac{1}{2}$: pass. 2 and -2: fail (both steep). 2 and $\frac{1}{2}$: fail (both uphill). The only pair that is the same size is 1 and -1.

✎ YOUR GO (pencil first)

Hold two pens crossed in a + shape and tilt the whole + slowly. One pen always points uphill while the other points downhill, and as one gets steeper the other gets flatter. That is the minus sign and the flip, in your hands.

PROCEDURE: parallel, perpendicular or neither?

1. Get each gradient: rearrange each equation to $y = mx + c$, or use the gradient formula on two points.
2. If $m_1 = m_2$: **parallel**. If $m_1 \times m_2 = -1$: **perpendicular**. Otherwise: **neither**.
3. Write the conclusion as a sentence that quotes the numbers (this is the "proof" an examiner wants).

Two-line questions are one-line questions in disguise

A question with two lines looks like twice the work. It isn't. Every one is one of **three kinds**, and each kind turns straight back into things you can already do with one line:

Kind	It sounds like	What you actually do
1. Compare	"Show that these lines are parallel." "Parallel, perpendicular or neither?" "Show the triangle has a right angle."	Find each line's m on its own (a one-line job, done twice). Then ask: are they equal? Do they multiply to -1?
2. Build	"Find the line through P parallel (or perpendicular) to L." "Find the perpendicular bisector."	Take m from the given line. Keep it (parallel), or flip it and change its sign (perpendicular). Now you have **one point and one gradient**: $y - y_1 = m(x - x_1)$.
3. Meet	"Where do the lines meet / cross / intersect?"	At the crossing point both equations are true, so set them equal and solve.

The big idea for Build questions: the line you are given is only a **gradient donor**. Take its m and ignore the rest of it, especially its c: your new line goes through a different point, so it gets its own c.

Decoder: what the scary words tell you to do

When the question says...	...it means
"parallel to"	use the **same** m
"perpendicular to", "at right angles to", "normal to", $\perp$	flip m and change its sign: $-\frac{1}{m}$
"passes through (a, b)", "(a, b) lies on"	put $x = a$ and $y = b$ into the equation

When the question says...	...it means
"meet", "cross", "intersect", "point of intersection"	set the two equations equal
"bisect", "bisector"	cut exactly in half: you need the **midpoint**
"show that", "prove"	do the working, then finish with one sentence that quotes the numbers

Example: L_1 has equation $4x + 2y = 5$. L_2 passes through $(1, 4)$ and is perpendicular to L_1. Find the equation of L_2.

Translate it into words first. It's a **Build**. L_1 is only the gradient donor. "Perpendicular" means flip and change the sign. "Passes through (1, 4)" is my point.

$$2y = -4x + 5 \Rightarrow y = -2x + 2.5$$ ← donor: $m_1 = -2$ (its c is not needed)

$$m_2 = +\frac{1}{2}$$ ← flip 2 to ½, and − becomes +

$$y - 4 = \frac{1}{2}(x - 1) \Rightarrow y = \frac{1}{2}x + 3.5$$ ← point (1, 4), gradient ½

Sense check: $-2 \times \frac{1}{2} = -1$ ✓. One line downhill, one uphill; one steep, one gentle ✓. Does (1, 4) fit? $\frac{1}{2} + 3.5 = 4$ ✓

Proving lines are parallel or perpendicular

Example: show that $y = 3x - 2$ and $6x - 2y = 7$ are parallel.

$$6x - 2y = 7 \;\Rightarrow\; 2y = 6x - 7 \;\Rightarrow\; y = 3x - 3.5$$

rearrange the second line

$$m_1 = 3, \quad m_2 = 3 \longleftarrow$$ *read the gradients*

Conclusion: both lines have gradient 3 (and different intercepts), so they are parallel.

Example: are $2x + 3y = 6$ and $3x - 2y = 4$ parallel, perpendicular or neither?

$$3y = -2x + 6 \;\Rightarrow\; y = -\tfrac{2}{3}x + 2 \longleftarrow$$ $m_1 = -2/3$

$$-2y = -3x + 4 \;\Rightarrow\; y = \tfrac{3}{2}x - 2 \longleftarrow$$ $m_2 = 3/2$ (÷ −2 flips every sign)

$$m_1 m_2 = -\tfrac{2}{3} \times \tfrac{3}{2} = -1$$

Conclusion: the product of the gradients is -1, so the lines are perpendicular.

Example: $y = 2x + 1$ and $y = -2x + 5$.

$m_1 = 2$, $m_2 = -2$. Not equal, and $2 \times (-2) = -4 \neq -1$. **Neither**: they cross, but not at a right angle. (**Changing only the sign is not enough** for perpendicular; you must also flip.)

Example: A(1, 2), B(4, 8), C(0, 5), D(6, 2). Show AB is perpendicular to CD.

$$m_{AB} = \frac{8 - 2}{4 - 1} = \frac{6}{3} = 2$$

$$m_{CD} = \frac{2 - 5}{6 - 0} = \frac{-3}{6} = -\tfrac{1}{2}$$

$$m_{AB} \times m_{CD} = 2 \times \left(-\tfrac{1}{2}\right) = -1 \longleftarrow$$ *so AB ⊥ CD*

Example: show the triangle A(1, 1), B(3, 5), C(7, 3) has a right angle at B.

$$m_{AB} = \frac{5-1}{3-1} = 2, \qquad m_{BC} = \frac{3-5}{7-3} = -\tfrac{1}{2}$$

Conclusion: $2 \times (-\frac{1}{2}) = -1$, so AB is perpendicular to BC: the angle at B is 90°.

Parallel or perpendicular lines through a point

PROCEDURE: line parallel / perpendicular to a given line, through a given point

1. Find the gradient of the given line (rearrange to $y = mx + c$ if needed).
2. Parallel: use the **same** gradient. Perpendicular: use $-\frac{1}{m}$.
3. Use $y - y_1 = m(x - x_1)$ with the given point, then tidy into $y = mx + c$.

Example: the line parallel to $y = 3x - 1$ through (2, 9).

$$m = 3: \;\; y - 9 = 3(x - 2) \;\Rightarrow\; y = 3x + 3$$

same gradient

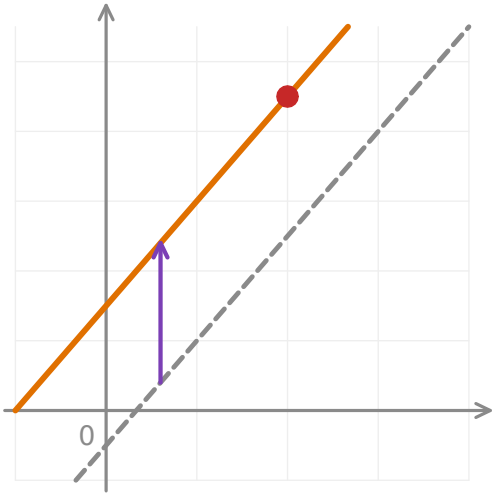
Example: the line perpendicular to $2x + 3y = 6$ through (4, 1).

$$y = -\tfrac{2}{3}x + 2 \;\Rightarrow\; m = -\tfrac{2}{3}$$ ← *rearrange the given line*

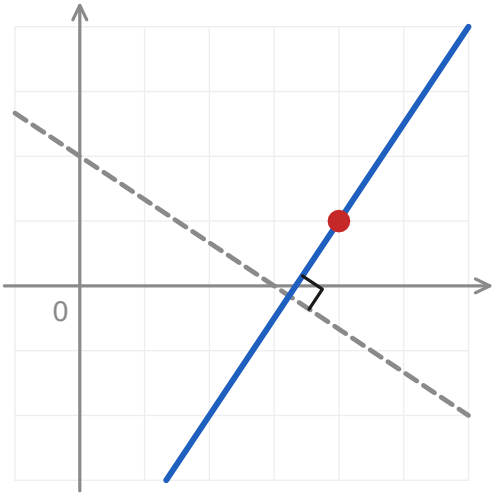
$$m_\perp = -\frac{1}{-\frac{2}{3}} = \tfrac{3}{2}$$ ← *negative reciprocal*

$$y - 1 = \tfrac{3}{2}(x - 4) \;\Rightarrow\; y = \tfrac{3}{2}x - 5$$

What those two answers look like (grey dashed = the given line, red dot = the given point):



Parallel: same tilt. Slide it up until it hits (2, 9).



Perpendicular: turn it a quarter turn, then slide it through (4, 1).

RULE: perpendicular bisector of AB

The line that cuts AB in half at a right angle. It passes through the **midpoint** of AB with gradient $-\frac{1}{m_{AB}}$.

Example: the perpendicular bisector of A(2, 1) and B(6, 5).

$$M = \left(\tfrac{2+6}{2}, \tfrac{1+5}{2}\right) = (4, 3)$$ ← *midpoint (next section)*

$$m_{AB} = \frac{5-1}{6-2} = 1 \;\Rightarrow\; m_{\perp} = -1$$

$$y - 3 = -(x - 4) \;\Rightarrow\; y = -x + 7$$

Midpoint and distance

RULE: midpoint and distance between two points

Midpoint of (x_1, y_1) and (x_2, y_2): average the xs and average the ys:

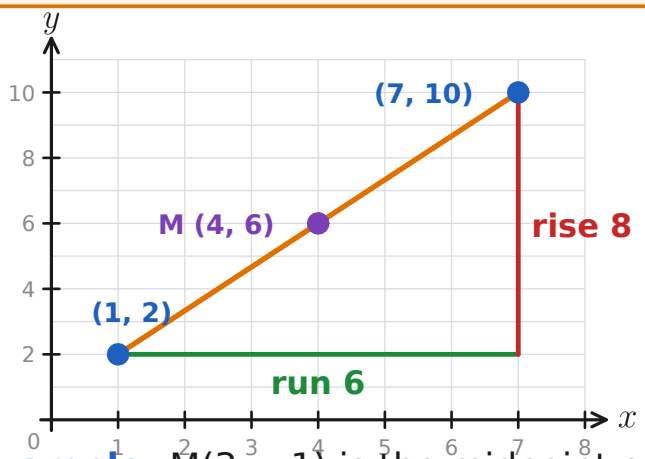
$$M = \left(\frac{x_1 + x_2}{2},\ \frac{y_1 + y_2}{2}\right).$$

Distance: the run and rise are the two short sides of a right-angled triangle, so by Pythagoras ($a^2 + b^2 = c^2$, Chapter 28):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Missing endpoint: if M is the midpoint of A and B, then

$$B = (2x_M - x_A,\ 2y_M - y_A).$$



midpoint: ((1 + 7)/2, (2 + 10)/2)
= (4, 6)

distance: run 6, rise 8

√(6² + 8²) = √100 = 10

Example: M(3, −1) is the midpoint of A(1, 4) and B. Find B.

$$\frac{1+x_B}{2} = 3 \Rightarrow x_B = 5; \quad \frac{4+y_B}{2} = -1 \Rightarrow y_B = -6$$

So $B = (5, -6)$. Check: the midpoint of (1, 4) and (5, −6) is (3, −1) ✓

Where two lines cross

RULE: intersection of two lines

At the crossing point both equations are true at once, so solve them as **simultaneous equations** (Chapter 15).

If both start $y =$, set the right-hand sides equal and solve for x. Otherwise substitute one into the other.

Put x back into either equation to get y. The answer is a **point** (x, y).

Parallel lines (same m, different c) never cross: the equation has no solution.

Example: where do $y = 2x - 4$ and $y = -x + 5$ meet?

$$2x - 4 = -x + 5$$ ← both equal y, so equal each other

$$3x = 9 \;\Rightarrow\; x = 3$$ ← +x and +4 both sides

$$y = 2(3) - 4 = 2$$ ← back into the first line

They meet at **(3, 2)**. Check in the second line: $-3 + 5 = 2$ ✓

Example: $2x + 3y = 12$ and $y = x - 1$.

$$2x + 3(x - 1) = 12 \;\Rightarrow\; 5x = 15 \;\Rightarrow\; x = 3,\; y = 2$$

substitute y = x − 1

Find the unknown: partial-information problems

Exam questions often give you **part** of the information about a line and a letter (often k, a or p) for something missing. They are all solved the same way:

PROCEDURE: partial-information problems

1. Write down what each piece of information means as an equation: a point on the line means its coordinates satisfy the equation; a given gradient means the gradient formula equals that number; parallel or perpendicular gives you m.
2. Solve the equation for the unknown.
3. Check by substituting back.

Example: the point $(k, 7)$ lies on $y = 2x - 3$. Find k.

$$7 = 2k - 3 \;\Rightarrow\; 2k = 10 \;\Rightarrow\; k = 5$$ ← put x = k, y = 7

Example: does (3, 5) lie on $y = 2x - 1$? Does (4, 6)?

$2(3) - 1 = 5$ ✓ so (3, 5) is on the line. $2(4) - 1 = 7 \neq 6$, so (4, 6) is not.

Example: the gradient from $(2, k)$ to $(5, 11)$ is 3. Find k.

$$\frac{11 - k}{5 - 2} = 3 \;\Rightarrow\; 11 - k = 9 \;\Rightarrow\; k = 2$$ ← gradient formula = 3

Example: $y = kx + 4$ is parallel to $6x - 2y = 1$. Find k. What if it were perpendicular?

$$y = 3x - \tfrac{1}{2} \Rightarrow m = 3$$ ← rearrange the given line

Parallel: $k = 3$. Perpendicular: $k = -\frac{1}{3}$.

Example: a line passes through (1, 3) and has y-intercept 5. Find its gradient.

$$3 = m(1) + 5 \;\Rightarrow\; m = -2$$ ← c = 5; use (1, 3)

Example: the line $ax + 2y = 8$ passes through (2, 1). Find a.

$$a(2) + 2(1) = 8 \;\Rightarrow\; a = 3$$

RULE: collinear points

Three points A, B, C are **collinear** (on one straight line) when $m_{AB} = m_{BC}$: the same gradient, and they share the point B.

Example: are A(1, 1), B(3, 5), C(6, 11) collinear?

$$m_{AB} = \frac{4}{2} = 2, \quad m_{BC} = \frac{6}{3} = 2$$ ← *equal, so yes: collinear*

Example: find k so that (1, 1), (3, 5) and (5, k) are collinear.

$$\frac{k-5}{5-3} = 2 \Rightarrow k = 9$$ ← *gradient must also be 2*

Mistakes to avoid

✗ Common mistakes

✗ $m = \dfrac{x_2 - x_1}{y_2 - y_1}$ ✓ $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

rise over run: the y's go on top

✗ $m = \dfrac{12-4}{4-8}$ ✓ $m = \dfrac{12-4}{8-4}$

same order top and bottom

✗ $2x + 3y = 6 : \ m = 2$ ✓ $y = -\frac{2}{3}x + 2 : \ m = -\frac{2}{3}$

rearrange to y = mx + c before reading m

✗ $m = 3 \Rightarrow m_\perp = -3$ ✓ $m = 3 \Rightarrow m_\perp = -\frac{1}{3}$

flip AND change sign

✗ $(-3, -2) : \ y - 2 = m(x - 3)$ ✓ $(-3, -2) : \ y + 2 = m(x + 3)$

subtracting a negative gives +

Straight lines in real life: temperature

Fahrenheit and Celsius are linked by a straight line: $F = 1.8C + 32$. Water freezes at 0 °C, which is $c = 32$ °F; every 1 °C warmer is 1.8 °F warmer.

$$F = 1.8(100) + 32 = 212$$ ← *boiling point: 100 °C = 212 °F ✓*

$$C = \frac{F - 32}{1.8}$$ ← *undo it for the reverse conversion (Chapter 14)*

Any time a question says "a fixed amount plus so much per...", think $y = mx + c$.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Straight line	$y = mx + c$ m = gradient, c = y-intercept; rearrange to this form first
Gradient	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$ same order top and bottom
Point-gradient form	$y - y_1 = m(x - x_1)$ one point and the gradient
Horizontal / vertical	$y = b \quad (m = 0), \qquad x = a \quad (m \text{ undefined})$
Parallel	$m_1 = m_2$
Perpendicular	$m_1 m_2 = -1, \qquad m_2 = -\dfrac{1}{m_1}$ except horizontal with vertical
Perpendicular sense check	one uphill and one downhill; one steep and one gentle
Two-line questions	Compare (find both m's), Build (borrow m, add a point), Meet (set equal)
Midpoint	$\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$
Distance	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Axis crossings	put $x = 0$ for the y-intercept; put $y = 0$ for the x-intercept
Intersection	solve the two equations simultaneously
Point on a line	its coordinates make the equation true
Collinear	$m_{AB} = m_{BC}$
Perpendicular bisector	through the midpoint, gradient $-1/m_{AB}$

Your turn

Answers in grey; worked solutions for 12 onwards on the next page.

Reading y = mx + c

1. Gradient and intercept of $y = 5x - 2$?
2. Gradient and intercept of $y = 7 - 3x$?
3. Gradient and intercept of $4x + 2y = 10$?
4. A plumber charges £40 call-out plus £30 per hour. Write the cost y for x hours.

1. m = 5, c = −2 2. m = −3, c = 7 3. y = −2x + 5: m = −2, c = 5 4. y = 30x + 40

Two points

5. Gradient between (1, 2) and (4, 11)
6. Equation through (1, 2) and (4, 11)
7. Equation through (−2, 7) and (2, −1)
8. Equation through (0, 5) parallel to $y = 3x - 1$

5. 3 6. y = 3x − 1 7. y = −2x + 3 8. y = 3x + 5

Sketch and perpendicular

9. Where does $3x + 5y = 15$ cross the axes?
10. Perpendicular gradient to m = 4?
11. Perpendicular gradient to m = −2/3?

9. (0, 3) and (5, 0) 10. −1/4 11. 3/2

Parallel, perpendicular or neither?

12. $y = 4x + 1$ and $y = 4x - 7$
13. $y = 2x + 3$ and $x + 2y = 6$
14. $3x + y = 2$ and $y = 3x + 1$
15. A(0, 1), B(2, 7), C(1, 5), D(4, 4): AB and CD

12. parallel 13. perpendicular 14. neither 15. perpendicular

Lines through a point

16. Parallel to $y = 2x + 7$ through (3, 1)
17. Perpendicular to $y = 2x + 7$ through (4, 0)
18. Perpendicular bisector of (0, 0) and (4, 8)

16. y = 2x − 5 17. y = −x/2 + 2 18. y = −x/2 + 5

Midpoint, distance, intersection

19. Midpoint of $(-2, 3)$ and $(6, 9)$
20. Distance from $(1, 1)$ to $(4, 5)$
21. $M(2, 5)$ is the midpoint of $A(-1, 3)$ and B. Find B.
22. Where do $y = 3x - 2$ and $y = x + 4$ meet?

19. $(2, 6)$ 20. 5 21. $(5, 7)$ 22. $(3, 7)$

Find the unknown

23. $(k, 11)$ lies on $y = 3x - 4$. Find k.
24. The gradient from $(1, k)$ to $(4, 10)$ is 2. Find k.
25. $y = kx - 3$ is perpendicular to $y = 4x + 1$. Find k.
26. $ax + 3y = 12$ passes through $(3, 2)$. Find a.
27. Are $(0, 2)$, $(2, 8)$, $(5, 17)$ collinear?
28. Gradient -2, through $(3, 1)$: equation, and where it crosses the x-axis?

23. $k = 5$ 24. $k = 4$ 25. $k = -1/4$ 26. $a = 2$ 27. yes 28. $y = -2x + 7$; $(3.5, 0)$

Worked solutions, 12 to 28

12. $m_1 = 4 = m_2$ *parallel*

13. $x + 2y = 6 \Rightarrow y = -\frac{1}{2}x + 3;\quad 2 \times (-\frac{1}{2}) = -1$ *perpendicular*

14. $y = -3x + 2;\quad -3 \times 3 = -9$ *neither: not equal, product not –1*

15. $m_{AB} = \frac{6}{2} = 3,\ m_{CD} = \frac{-1}{3};\quad 3 \times (-\frac{1}{3}) = -1$ *perpendicular*

16. $y - 1 = 2(x - 3) \Rightarrow y = 2x - 5$

17. $m_\perp = -\frac{1}{2};\ y - 0 = -\frac{1}{2}(x - 4) \Rightarrow y = -\frac{1}{2}x + 2$

18. $M = (2, 4),\ m = 2,\ m_\perp = -\frac{1}{2};\ y - 4 = -\frac{1}{2}(x - 2) \Rightarrow y = -\frac{1}{2}x + 5$

19. $\left(\frac{-2+6}{2}, \frac{3+9}{2}\right) = (2, 6)$

20. $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$

21. $B = (2 \times 2 - (-1),\ 2 \times 5 - 3) = (5, 7)$

22. $3x - 2 = x + 4 \Rightarrow x = 3,\ y = 7$

23. $11 = 3k - 4 \Rightarrow k = 5$

24. $\frac{10-k}{3} = 2 \Rightarrow k = 4$

25. $4k = -1 \Rightarrow k = -\frac{1}{4}$ *perpendicular: product –1*

26. $3a + 6 = 12 \Rightarrow a = 2$

27. $\frac{8-2}{2} = 3,\ \frac{17-8}{3} = 3$ *equal gradients: collinear*

28. $1 = -6 + c \Rightarrow y = -2x + 7;\quad 0 = -2x + 7 \Rightarrow x = 3.5$